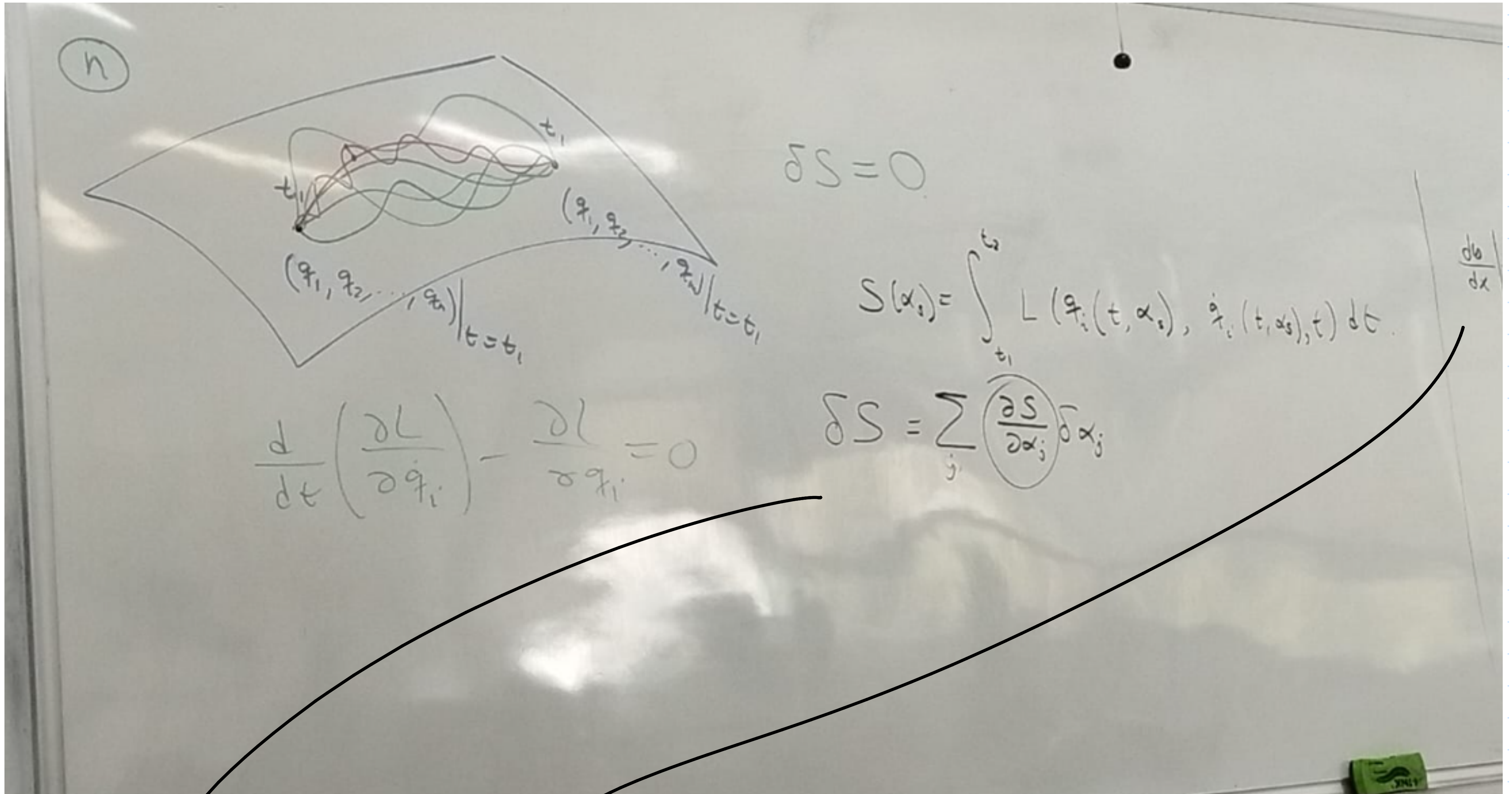


04/09/2025

Clase # 15

Mecánica teórica



$$\left. \frac{dy}{dx} \right|_{x=x_0} = \lim_{x \rightarrow x_0} \frac{y(x) - y(x_0)}{x - x_0}$$

Tenemos

$$\frac{\partial S}{\partial \alpha_j} = \sum_k \left[\frac{\partial L}{\partial q_k} \frac{\partial q_k}{\partial \alpha_j} + \frac{\partial L}{\partial \dot{q}_k} \frac{\partial \dot{q}_k}{\partial \alpha_j} \right]$$

$$\frac{\partial^2 q_k}{\partial \alpha_j \partial t} = \frac{\partial}{\partial t} \left(\frac{\partial q_k}{\partial \alpha_j} \right)$$

Tenemos

$$\frac{\partial S}{\partial \alpha_j} = \sum_k \left[\frac{\partial L}{\partial q_k} \frac{\partial q_k}{\partial \alpha_j} + \frac{\partial L}{\partial \dot{q}_k} \frac{\partial}{\partial t} \left(\frac{\partial q_k}{\partial \alpha_j} \right) \right]$$

Nos fijamos en:

$$* \frac{d}{dt} \left[\left(\frac{\partial L}{\partial \dot{q}_k} \right) \left(\frac{\partial q_k}{\partial \alpha_j} \right) \right] = \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] \frac{\partial q_k}{\partial \alpha_j} + \left(\frac{\partial L}{\partial \dot{q}_k} \right)$$

$$\frac{d}{dt} \left(\frac{\partial q_k}{\partial \alpha_j} \right)$$



$$\frac{\partial}{\partial t} \left(\frac{\partial q_k}{\partial \alpha_j} \right)$$

The image shows a handwritten derivation on a piece of paper. At the top, it states:

$$\frac{\partial S}{\partial \alpha_j} = \sum_k \left[\frac{\partial L}{\partial q_k} \frac{\partial q_k}{\partial \alpha_j} + \left(\frac{\partial L}{\partial \dot{q}_k} \frac{\partial}{\partial t} \left(\frac{\partial q_k}{\partial \alpha_j} \right) \right) \right]$$

A horizontal line is drawn below this equation. Below the line, it starts with an asterisk and shows the derivation of the time derivative term:

$$* \frac{d}{dt} \left[\left(\frac{\partial L}{\partial \dot{q}_k} \right) \left(\frac{\partial q_k}{\partial \alpha_j} \right) \right] = \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] \frac{\partial q_k}{\partial \alpha_j} + \left(\frac{\partial L}{\partial \dot{q}_k} \right) \frac{d}{dt} \left(\frac{\partial q_k}{\partial \alpha_j} \right)$$

The last term is further simplified to:

$$\frac{\partial}{\partial t} \left(\frac{\partial q_k}{\partial \alpha_j} \right)$$

Tenemos ...

$$\frac{\partial S}{\partial \alpha_j} = \sum_k \left[\frac{\partial L}{\partial q_k} \frac{\partial q_k}{\partial \alpha_j} + \frac{d}{dt} \left\{ \left(\frac{\partial L}{\partial \dot{q}_k} \right) \left(\frac{\partial q_k}{\partial \alpha_j} \right) \right\} \right] -$$

$$\left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right) \right] \frac{\partial q_k}{\partial \alpha_j}$$

Si solo tenemos un sólo grado de libertad.

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad ; \quad q = q(t, \alpha)$$

Tenemos

$$S(\alpha) = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} \frac{\partial q}{\partial \alpha} + \underbrace{\frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial \alpha}} \right] dt$$

Tenemos

$$\frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial \alpha} = \frac{\partial L}{\partial \dot{q}} \frac{\partial}{\partial \alpha} \left(\frac{dq}{dt} \right) = \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} \left(\frac{\partial q}{\partial \alpha} \right)$$

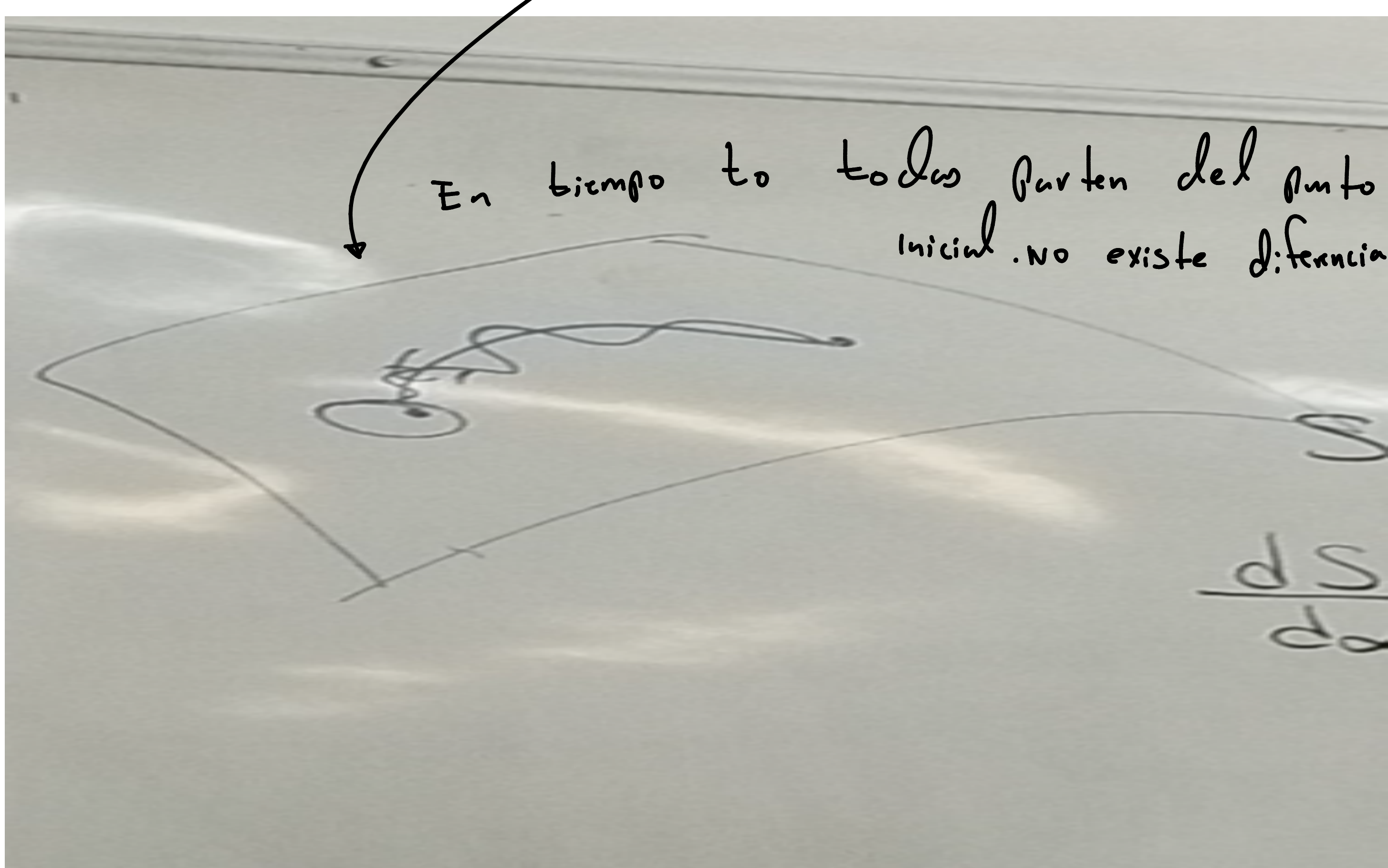
$$= \frac{d}{dt} \left[\left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] - \left\{ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right\} \left(\frac{\partial q}{\partial \alpha} \right)$$

Tenemos ...

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial q}{\partial \dot{q}} \right) &= \frac{\partial}{\partial \dot{q}} \left(\frac{\partial q}{\partial t} \right) \\ &= \frac{d}{dt} \left[\left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] - \left\{ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right\} \left(\frac{\partial q}{\partial \alpha} \right) \\ &= \int_{t_1}^{t_2} \left\{ \frac{\partial L}{\partial q} + \frac{\partial}{\partial \alpha} \left[\left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] - \left[\left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] \frac{d}{dt} \left(\frac{\partial q}{\partial \alpha} \right) \right\} dt \end{aligned}$$

$$\begin{aligned} S &= \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \\ \frac{dS}{d\alpha} &= \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] dt \\ &\quad + \left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \Big|_{t_1}^{t_2} \\ &= \int_{t_1}^{t_2} \left\{ \frac{\partial L}{\partial q} + \frac{\partial}{\partial \alpha} \left[\left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] - \left[\left(\frac{\partial L}{\partial \dot{q}} \right) \left(\frac{\partial q}{\partial \alpha} \right) \right] \frac{d}{dt} \left(\frac{\partial q}{\partial \alpha} \right) \right\} dt \end{aligned}$$

$$\frac{\partial q(t, \alpha)}{\partial \alpha} \Big|_{t_1} = 0 \quad (\text{Explicación})$$



Tenemos

$$\frac{ds}{d\alpha} = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \left(\frac{\partial q}{\partial \alpha} \right) dt$$

Tenemos...

$$\begin{aligned} \frac{ds}{d\alpha} &= \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \left(\frac{\partial q}{\partial \alpha} \right) dt \\ \delta S &= \left(\frac{ds}{d\alpha} \right)_{\alpha=0} \delta \alpha = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \left(\frac{\partial q}{\partial \alpha} \right)_{\alpha=0} \delta \alpha dt \\ &= \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right]_{\alpha=0} \delta q dt \end{aligned}$$

$\delta q = \left(\frac{\partial q}{\partial \alpha} \right)_{\alpha=0} \delta \alpha$

$$\delta S = \left(\frac{dS}{d\alpha} \right)_{\alpha=0} \delta\alpha = \int_{t_1}^{t_2} \left\{ \frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right\} \delta q dt = 0$$

$$\delta q = \left(\frac{\partial q}{\partial \alpha} \right)_{\alpha=0} \delta\alpha$$

Ecuaciones de
Euler-Lagrange

Unidades de la acción $\left[\frac{\text{Energía}}{\text{tiempo}} \right] \rightarrow$ mismas unidades que la constante de Planck.

Checkar dem c/ n grados de libertad.

Revisar problema $\rightarrow$ ¿Qué pasa con la acción?

$$S = \int_{t_1}^{t_2} L dt \quad \left. \vphantom{\int_{t_1}^{t_2}} \right\} \text{ De acá se obtienen las EC. } \underline{\text{E-L}}$$

• Idea $\rightarrow$ calcular la acción para partícula libre

$$L = \frac{1}{2} m \dot{\vec{r}}^2$$

$\vec{r}(t) = \vec{r}_0 + \vec{v}_0 t$ } Martes (Evaluando para el caso de partícula libre)